

## SUPPLEMENTARY RESULTS: HiPAMA

This document contains the model-specific component and loss-function ablations for HiPAMA, accompanying *Beyond the Majority Bin: A Methodological Framework for Balanced Automatic Pronunciation Assessment*. The cross-model comparisons and experimental reproducibility settings are reported in the paper’s supplementary material.

### I. READING GUIDE

Baseline denotes the original model and training procedure. The +CV configuration introduces speaker-independent grouped cross-validation using `GroupKFold(n_splits=5)` without shuffling. Both +DB and +MS build on CV: +DB adds Dynamic Balanced MSE, whereas +MS adds Macro-MSE-based checkpoint selection. Complete BOMA combines CV, DB, and MS. In the loss-function ablations, CV and MS are held fixed while conventional MSE, DB-MSE, and Score-Balanced Loss (SB) are compared.

CCC-BS measures distribution-balanced agreement and PCC measures linear association; higher values are preferred for both. Micro-MSE and Macro-MSE are error measures on the unified  $[0, 10]$  score scale, so lower values are preferred. Boldface marks the best value in each comparison. These metrics express complementary evaluation objectives; improvements under balanced criteria need not coincide with improvements under sample-weighted criteria.

### II. RESULTS

HiPAMA shows one of the clearest balanced-performance gains: BOMA is best on eight of nine CCC-BS aspects, with +DB remaining best only for utterance completeness (Table S1). BOMA also attains the lowest Macro-MSE for all nine aspects, although the lowest Micro-MSE is usually found under the Baseline or CV configurations (Table S2). PCC follows a different pattern, favoring the Baseline on seven aspects and CV on two (Table S3). When only the loss is changed, DB-MSE is best on eight CCC-BS aspects and all nine Macro-MSE aspects (Tables S4 and S5).

TABLE S1  
ABLATION STUDY FOR HiPAMA. CORRELATIONS ARE MEASURED USING CCC-BS.

| Configuration | Phone Score  | Word Score (CCC-BS) |              |              | Utterance Score (CCC-BS) |              |              |              |              |
|---------------|--------------|---------------------|--------------|--------------|--------------------------|--------------|--------------|--------------|--------------|
|               | Phoneme ↑    | Acc. ↑              | Stress ↑     | Total ↑      | Acc. ↑                   | Comp. ↑      | Fluency ↑    | Prosody ↑    | Total ↑      |
| Baseline      | 0.414        | 0.496               | 0.110        | 0.538        | 0.690                    | 0.018        | 0.686        | 0.695        | 0.745        |
| + CV          | 0.400        | 0.507               | 0.114        | 0.583        | 0.702                    | 0.018        | 0.702        | 0.711        | 0.758        |
| + DB          | 0.537        | 0.639               | 0.535        | 0.561        | 0.725                    | <b>0.059</b> | 0.724        | 0.700        | 0.758        |
| + MS          | 0.429        | 0.531               | 0.122        | 0.582        | 0.692                    | 0.043        | 0.674        | 0.678        | 0.729        |
| <b>BOMA</b>   | <b>0.579</b> | <b>0.641</b>        | <b>0.583</b> | <b>0.597</b> | <b>0.737</b>             | 0.041        | <b>0.750</b> | <b>0.735</b> | <b>0.779</b> |

<sup>a</sup> The +DB and +MS configurations are evaluated on top of the CV configuration; BOMA combines CV, DB, and MS.

<sup>b</sup> Bold values indicate the highest CCC-BS value for each aspect within this ablation.

TABLE S2  
MICRO-MSE AND MACRO-MSE ABLATION STUDY FOR HiPAMA. LOWER VALUES INDICATE BETTER PERFORMANCE.

| Configuration | Phone        |              | Word Score   |              |              |              |              |              | Utterance Score |              |              |               |              |              |              |              |              |              |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-----------------|--------------|--------------|---------------|--------------|--------------|--------------|--------------|--------------|--------------|
|               | Phoneme      |              | Acc.         |              | Stress       |              | Total        |              | Acc.            |              | Comp.        |               | Fluency      |              | Prosody      |              | Total        |              |
|               | Mic. ↓       | Mac. ↓       | Mic. ↓       | Mac. ↓       | Mic. ↓       | Mac. ↓       | Mic. ↓       | Mac. ↓       | Mic. ↓          | Mac. ↓       | Mic. ↓       | Mac. ↓        | Mic. ↓       | Mac. ↓       | Mic. ↓       | Mac. ↓       | Mic. ↓       | Mac. ↓       |
| Baseline      | <b>2.093</b> | 16.731       | <b>2.151</b> | 12.042       | <b>0.178</b> | 10.129       | <b>1.630</b> | 10.988       | <b>1.126</b>    | 3.355        | 0.259        | 28.135        | <b>0.914</b> | 5.002        | 0.931        | 4.810        | <b>1.057</b> | 4.023        |
| + CV          | 2.208        | 17.007       | 2.204        | 11.727       | 0.181        | 9.954        | 1.678        | 9.799        | 1.144           | 3.269        | 0.258        | 28.262        | 0.930        | 4.831        | <b>0.929</b> | 4.606        | 1.067        | 3.834        |
| + DB          | 6.425        | 7.969        | 8.217        | 4.895        | 1.704        | 4.091        | 5.116        | 6.462        | 1.406           | 2.630        | 0.249        | 25.747        | 1.071        | 4.473        | 1.073        | 4.606        | 1.293        | 3.658        |
| + MS          | 2.455        | 15.348       | 2.382        | 10.671       | 0.191        | 9.666        | 1.812        | 9.263        | 1.211           | 2.789        | 0.248        | 26.657        | 0.986        | 5.125        | 0.973        | 4.911        | 1.140        | 3.944        |
| <b>BOMA</b>   | 8.003        | <b>6.689</b> | 9.892        | <b>4.711</b> | 2.212        | <b>3.781</b> | 6.079        | <b>5.381</b> | 1.769           | <b>2.182</b> | <b>0.245</b> | <b>25.495</b> | 1.289        | <b>3.799</b> | 1.216        | <b>3.823</b> | 1.593        | <b>3.132</b> |

<sup>a</sup> Micro-MSE (Mic.) and Macro-MSE (Mac.) are reported on the unified  $[0, 10]$  scale; lower values (↓) indicate better performance.

<sup>b</sup> The +DB and +MS configurations are evaluated on top of CV; BOMA combines CV, DB, and MS.

<sup>c</sup> Bold values indicate the lowest error for each metric within the model-specific ablation.

TABLE S3  
PCC ABLATION STUDY FOR HIPAMA. HIGHER VALUES INDICATE BETTER PERFORMANCE.

| Configuration | Phone Score  | Word Score (PCC) |              |              | Utterance Score (PCC) |              |              |              |              |
|---------------|--------------|------------------|--------------|--------------|-----------------------|--------------|--------------|--------------|--------------|
|               | Phoneme ↑    | Acc. ↑           | Stress ↑     | Total ↑      | Acc. ↑                | Comp. ↑      | Fluency ↑    | Prosody ↑    | Total ↑      |
| Baseline      | <b>0.616</b> | <b>0.575</b>     | 0.320        | <b>0.591</b> | <b>0.730</b>          | 0.276        | <b>0.749</b> | <b>0.751</b> | <b>0.754</b> |
| + CV          | 0.593        | 0.561            | <b>0.326</b> | 0.577        | 0.725                 | <b>0.357</b> | 0.742        | 0.744        | 0.749        |
| + DB          | 0.519        | 0.475            | 0.229        | 0.508        | 0.700                 | 0.299        | 0.720        | 0.728        | 0.728        |
| + MS          | 0.560        | 0.537            | 0.280        | 0.554        | 0.721                 | 0.325        | 0.738        | 0.742        | 0.743        |
| <b>BOMA</b>   | 0.497        | 0.453            | 0.209        | 0.482        | 0.690                 | 0.273        | 0.719        | 0.729        | 0.719        |

<sup>a</sup> The +DB and +MS configurations are evaluated on top of CV; BOMA combines CV, DB, and MS.

<sup>b</sup> Bold values indicate the highest PCC value for each aspect within the model-specific ablation.

TABLE S4  
LOSS-FUNCTION ABLATION IN TERMS OF CCC-BS FOR HIPAMA. ALL CONFIGURATIONS USE THE SAME CROSS-VALIDATION AND MACRO-MSE-BASED MODEL-SELECTION PROTOCOL; ONLY THE TRAINING LOSS IS CHANGED.

| Loss   | Phone Score  | Word Score (CCC-BS) |              |              | Utterance Score (CCC-BS) |              |              |              |              |
|--------|--------------|---------------------|--------------|--------------|--------------------------|--------------|--------------|--------------|--------------|
|        | Phoneme ↑    | Acc. ↑              | Stress ↑     | Total ↑      | Acc. ↑                   | Comp. ↑      | Fluency ↑    | Prosody ↑    | Total ↑      |
| MSE    | 0.429        | 0.531               | 0.122        | 0.582        | 0.692                    | <b>0.043</b> | 0.674        | 0.678        | 0.729        |
| DB-MSE | <b>0.579</b> | <b>0.641</b>        | <b>0.583</b> | <b>0.597</b> | <b>0.737</b>             | 0.041        | <b>0.750</b> | <b>0.735</b> | <b>0.779</b> |
| SB     | 0.376        | 0.506               | 0.118        | 0.573        | 0.658                    | 0.027        | 0.673        | 0.697        | 0.735        |

<sup>a</sup> MSE corresponds to the CV+MS configuration, i.e., the BOMA validation and model-selection protocol trained with conventional MSE.

<sup>b</sup> DB-MSE corresponds to the proposed Dynamic Balanced MSE used in the complete BOMA framework.

<sup>c</sup> SB denotes Score-Balanced Loss, which replaces DB-MSE while keeping the remaining protocol unchanged.

<sup>d</sup> Bold values indicate the highest CCC-BS for each aspect within the model-specific loss ablation.

TABLE S5  
LOSS-FUNCTION ABLATION IN TERMS OF MICRO-MSE AND MACRO-MSE FOR HIPAMA. ALL CONFIGURATIONS USE THE SAME CROSS-VALIDATION AND MACRO-MSE-BASED MODEL-SELECTION PROTOCOL; ONLY THE TRAINING LOSS IS CHANGED.

| Loss   | Phone   |        | Word Score |        |        |        |        |        | Utterance Score |        |        |        |         |        |         |        |        |        |
|--------|---------|--------|------------|--------|--------|--------|--------|--------|-----------------|--------|--------|--------|---------|--------|---------|--------|--------|--------|
|        | Phoneme |        | Acc.       |        | Stress |        | Total  |        | Acc.            |        | Comp.  |        | Fluency |        | Prosody |        | Total  |        |
|        | Mic. ↓  | Mac. ↓ | Mic. ↓     | Mac. ↓ | Mic. ↓ | Mac. ↓ | Mic. ↓ | Mac. ↓ | Mic. ↓          | Mac. ↓ | Mic. ↓ | Mac. ↓ | Mic. ↓  | Mac. ↓ | Mic. ↓  | Mac. ↓ | Mic. ↓ | Mac. ↓ |
| MSE    | 2.455   | 15.348 | 2.382      | 10.671 | 0.191  | 9.666  | 1.812  | 9.263  | 1.211           | 2.789  | 0.248  | 26.657 | 0.986   | 5.125  | 0.973   | 4.911  | 1.140  | 3.944  |
| DB-MSE | 8.003   | 6.689  | 9.892      | 4.711  | 2.212  | 3.781  | 6.079  | 5.381  | 1.769           | 2.182  | 0.245  | 25.495 | 1.289   | 3.799  | 1.216   | 3.823  | 1.593  | 3.132  |
| SB     | 2.561   | 17.189 | 2.483      | 11.313 | 0.182  | 9.803  | 1.934  | 9.710  | 1.180           | 3.111  | 0.253  | 27.472 | 0.978   | 5.236  | 0.965   | 4.624  | 1.161  | 3.955  |

<sup>a</sup> MSE corresponds to the CV+MS configuration, i.e., the BOMA validation and model-selection protocol trained with conventional MSE.

<sup>b</sup> DB-MSE corresponds to the proposed Dynamic Balanced MSE used in the complete BOMA framework.

<sup>c</sup> SB denotes Score-Balanced Loss, which replaces DB-MSE while keeping the remaining protocol unchanged.

<sup>d</sup> Micro-MSE (Mic.) and Macro-MSE (Mac.) are reported on the unified [0, 10] scale; lower values (↓) indicate better performance.

<sup>e</sup> Bold values indicate the lowest error for each metric within the model-specific loss ablation.